

GuruTej Pratap

Etawah, Uttar Pradesh 206001

+91 7037767875 · gurutej70@gmail.com · [linkedin.com/in/gurutejpratap](https://www.linkedin.com/in/gurutejpratap) · github.com/GuruTej-15

Training / Experience

Software Engineering Trainee | C++, Data Structures, MERN Stack, REST APIs

Jun 2026 - Jul 2026

Lovely Professional University

- Completed intensive training covering algorithmic complexity, linear/non-linear data structures, recursion, and searching/sorting.
- Engineered **Smart Blood Bank Management System** automating donor records, inventory across 8 blood groups, and emergency fulfillment.
- Implemented in-memory **Priority Queues & Binary Heaps ($O(\log n)$)**, **Hash Tables**, and secure **JWT RBAC** across 10+ REST endpoints.
- Live Demo:** <https://smart-blood-bank-management-system-topaz.vercel.app> · **Repository Link:** [GitHub](https://github.com)

Projects

RecordHub | Next.js 16, React 19, Node.js, Express.js, MongoDB, Tailwind CSS

Mar 2026 - Apr 2026

- Architected a full-stack platform using **Next.js 16 App Router & React 19** to track 10+ competition metrics, problem-solving history, and publish verified portfolios.
- Implemented multi-email authentication with **JWT** in **httpOnly** cookies, **bcrypt** hashing, **Google OAuth 2.0 SSO**, and 6-digit OTP reset via **NodeMailer**.
- Built full CRUD contest lifecycle tracking with **MongoDB/Mongoose** indexed schemas and aggregation pipelines, reducing profile load latency by **35%**.
- Live Demo:** <https://recordhub.vercel.app> · **Repository Link:** <https://github.com/GuruTej-15/recordhub>

OS File Locking & Concurrency Simulator | C, JavaScript (ES6+), Graph Algorithms (DFS), OS Synchronization

May 2026

- Engineered an OS synchronization simulator in **C** and **JavaScript** modeling **Readers-Writer locks** with FIFO queues, eliminating thread starvation across 15+ processes.
- Implemented **$O(V + E)$ two-level deadlock detection** using **DFS-based Resource Allocation Graph** cycle traversal with priority victim preemption.
- Built an automated stress-simulation engine across 100+ concurrent operations, optimizing victim heuristics to reduce abort rates by **40%**.
- Repository Link:** <https://github.com/GuruTej-15/os-file-locking-simulator>

Smart Flashcard & Interactive Quiz Engine | JavaScript (ES6+), PWA, Google Apps Script API, CSS3

Sep 2025

- Developed an interactive **Progressive Web App (PWA)** with **Service Workers** enabling 100% offline availability and sub-100ms deck load times with progress counters.
- Integrated **Google Apps Script & Google Sheets API** as a headless CMS to dynamically fetch and filter 50+ curated cards across multiple subject categories.
- Engineered **$O(n)$ Fisher-Yates shuffle** randomization, 3D CSS flip animations using **rotateY**, and regex parsers supporting 4+ MCQ formats with instant feedback.
- Live Demo:** <https://satgurunotes.netlify.app> · **Repository Link:** <https://github.com/GuruTej-15/Flashcards>

Certificates / Certifications

Oracle Certified Foundations Associate — Agentic AI | OCI AI

Aug 2026

Oracle Cloud Infrastructure — [Certificate Link \(Agentic AI\)](#) · [Certificate Link \(OCI AI\)](#)

Programming Using C++

Aug 2025

Infosys Springboard — [Certificate Link](#)

CyberSmart Awareness CSR Internship

Aug 2025

WNS Cares Foundation — [Certificate Link](#)

React.js Development

Mar 2025

Tech Veda — [Certificate Link](#)

Skills

Languages:

JavaScript (ES6+), C, C++, HTML5, CSS3, SQL

Technologies/Frameworks:

React.js, Next.js 16, Node.js, Express.js, Vite, Tailwind CSS, MongoDB, Mongoose, Recharts, Axios

Developer Tools & Cloud:

Git, GitHub Actions, Linux, Vercel, Postman, VS Code, Google Apps Script, npm

Domain Skills:

Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design

Education

Lovely Professional University

Aug 2024 - May 2028

Bachelor of Technology in Computer Science and Engineering — CGPA: 7.42

Phagwara, Punjab

Suditi Global Academy

2022 - 2023

Intermediate (Class XII) — Science (PCM) — Percentage: 64.8%

Etawah, Uttar Pradesh

Suditi Global Academy

2020 - 2021

Secondary School Examination (Class X) — Percentage: 74.8%

Etawah, Uttar Pradesh